2 It is given that

$$x = 1 + \frac{1}{t} \qquad \text{and} \qquad y = te^t.$$

(a) Show that $\frac{dy}{dx} = -e^t(t^3 + t^2)$. [3]

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(b) Find $\frac{d^2y}{dx^2}$ in terms of t. [4]

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